Tom Houlé

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Relevant Professional Experience

Rust Engineer Feb. 2026 – Present
Zed Industries Remote

Lead Engineer Oct. 2023 – Feb. 2026
Grafbase Remote

- Small, high ownership and high velocity team building a BaaS platform on Rust, Wasm and Cloudflare Workers, then an enterprise focused Federated GraphQL solution.
- Deep involvement in MCP and LLM routing solutions design.j
- Participated in the design and implementation of Grafbase Gateway, a GraphQL Federation Gateway that demonstrated best-in-class runtime and query planning performance. Co-developed the WebAssembly (WASI preview 2) based extensions system.
- Regular close collaboration with enterprise customers to shape and implement solutions for their enterprise GraphQL deployments. For example a revamp of the authentication and authorization architecture of the Grafbase platform for self-hosted scenarios, with deep integration with the enterprise's identity provider and policy engines (Cedar, AuthZEN).
- Led the design and development of schema evolution features, like schema proposals and usage-based (ClickHouse) breaking change analysis.
- Started and maintained the first and only Rust implementation of federated GraphQL composition.
- Engaged with GraphQL working groups (Composite Schemas) and gave talks at GraphQL Conf 2025.
- Owned the public Kubernetes (AWS EKS) deployment of the Grafbase platform: monitoring, integrations, OAuth/OIDC, …
- Designed and implemented a GraphQL gRPC translation layer in WebAssembly and a companion protoc plugin.

Senior Engineer, Tech Lead Sept. 2019 – Oct. 2023
Prisma Berlin, Remote

- Second Rust hire at Prisma. Started on the Scala to Rust migration (Prisma 1 to Prisma 2), then worked on the Rust components of the ORM over four years. Weekly downloads grew about 115x over that time frame.
- Three years as Tech Lead of the Schema Team, responsible for the Prisma Schema Language and its implementation, and the whole database schema management space.
- Subsequently, worked on the Client Team, responsible for the Query Engine behind Prisma Client.
- Frequently worked on database drivers and connection pools, using advanced debugging and profiling tools (Wireshark, debuggers, perf, dhat, tokio console, etc.). That lead to improved connection management, reduced cold start times on serverless, many fixed bugs and generally a better experience for users of the ORM.
- Owned the size project leading to considerable binary size reductions (in particular for serverless use cases), and spearheaded the WebAssembly migration effort.
- Reworked the Prisma Schema Language (PSL) implementation to use traditional programming language parsing techniques, massively improving performance and ease of implementing validations.
- Built the language intelligence infrastructure for the PSL LSP server and the PSL formatter.
- Introduced rigorous benchmarking in different dimensions (release sizes, compile time, resource usage, performance).
- Engaged in hiring for ICs, Engineering and Product Managers, mentoring and formal feedback rounds.
- Proposed and ran Rust and database engineering learning groups for engineers.
- Still all time top contributor to prisma-engines to this day.
- Led high impact projects: native types support, Postgres extensions, isolating the query planner, …

Engineer, Senior Engineer, Tech Lead Oct. 2016 – Sept. 2019
store2be Berlin

- Second engineering hire as Full Stack Engineer on a B2B booking platform with Ruby on Rails, Postgres and Kubernetes on the backend, and React on the frontend.
- Introduced and drove adoption of TypeScript, and later GraphQL to the team.
- As Tech Lead for the whole backend team, I was responsible for hiring, one-on-ones, career planning, formal feedback rounds, and mentoring.
- Bolstered collaboration practices: regular pair programming, systematic, thorough code reviews, regular 1:1s, TDD, product-oriented thinking.
- Built a PDF contract and invoice generation service with per-customer templates support using Rust with futures 0.1 and tokio.
- Took an active part in infrastructure work, migrating from Heroku to Kubernetes and Terraform.

Web Developer and Project Manager March 2015 – Oct. 2016
Médiéval AFDP Lyon

- Developed new internal web applications, new public-facing websites and enhanced older websites as the single full-time developer in a heritage-focused company working mostly for the public sector.
- Took on other tasks occasionally: web scraping, batch data manipulation…
- Python (Django), React, Typescript, PHP, PostgreSQL, Ansible, Jenkins, Docker.

Interests

- Long lived interest in **database engines** and database engineering in general.
- Long time **Rust** user who saw the language and ecosystem develop from the very early stages. I have been writing Rust code regularly since 2016–2017, and it has been my primary language at work since I joined Prisma in 2019. You can find more and less recent samples of my code on GitHub.
- Keen interest in **formal methods**. Decent working grasp of proof assistants (Lean, Coq, Agda) and the underlying theory. Not as knowledgeable, but interested in automatic verification.

Skills

- **Languages**: English, French, German, Chinese (HSK level 6 in 2014), Japanese (JLPT N1 in 2014)
- **Programming languages**: Rust, Zig, SQL, Lean, TypeScript, Python, Ruby, Scala, C, C++, etc.
- **Technologies**: Database Engineering, Linux, Kubernetes, Docker, Linux, DuckDB, GraphQL, etc.

Degrees

Master's Degree in Tourism Management Sept. 2014 – Sept. 2015
Université Lyon II

Master's Degree in Political Science Sept. 2008 – May 2013
Institut d'Études Politiques Aix-en-Provence